



Figure 1: Rockstor dashboard screenshot

and intuitive UI. Unlike several other NAS distros like FreeNAS, Open Media Vault, NAS for free, or even NAS boxes supporting BTRFS such as Synology or Netgear Ready NAS, Rockstor is the only distro that is purely based on Linux (CentOS) and BTRFS, and is fully open source under GPLv2. Also, compared to available public cloud storage options such as Google Drive, Dropbox, etc, storing data on Rockstor is safe, secure and cost-effective. Your data lives on your devices and on your premises, and you can easily expand or reduce storage capacity by adding or removing disks from the Web UI. You can even create a hybrid workflow, combining Rockstor with public cloud storage to have versatile access to your data.

Advanced Rockstor features

Rockstor makes it easy to manage advanced BTRFS features. It offers many advantages, such as mixing and matching drives of different types and sizes (you can combine your HDDs with USBs, external hard drives, etc), adding and removing disks, and easily switching between different RAID profiles.

Rockstor builds on top of Copy on Write (CoW) features of BTRFS to provide capabilities such as Snapshots (on demand or on schedule), Clones, appliance-to-appliance replication, to name a few. Snapshots can be conveniently created, cloned, scheduled and accessed by clients using NFS or Samba. Appliance management tasks such as appliance-to-appliance replication, adding a new appliance and making appliances aware of each other can be easily done.

All of the above operations can be carried out easily over the Web UI. So, with the combination of the Web UI and the command line, users with little IT experience as well as advanced users, and everyone else in between, can easily use Rockstor.

Installation

Installing Rockstor is straightforward and very similar to CentOS or Fedora. The Quickstart guide, available with Rockstor documentation, is a good place to start. The website (<http://rockstor.com>) gives the download instructions. For this tutorial, I have used my production

Rockstor system that runs on a HP ProLiant Microserver with four disks, and runs Rockstor version 3.8-9. You could use Oracle VirtualBox or other hypervisors like KVM, VMware, Xen, etc, for installation. If you install Rockstor on your preferred hardware (bare metal), then you will need either a CD or a bootable USB (you can either create your own or buy one from the Rockstor shop).

The Snapshot feature of Rockstor

A snapshot is the point-in-time state of a share (folder). Snapshot technology in general, and snapshot implementation of Copy-on-Write (CoW) in BTRFS (Linux) in particular, offers users simple data backup, data protection and data recovery tools.

Some possible real world scenarios that could use Snapshot technology

Imagine you are working on an important project such as writing a book or a grant proposal, editing a video you shot during your vacation, recording a podcast, editing your photos, creating a presentation, writing code or anything else of significance. You could encounter a number of problems while doing so:

1. You create your work (say four paragraphs of a proposal or a video clip), and save it. You then edit and save it again (say, you delete a paragraph or remove some video footage). Then, for some reason, you want to compare your work and need to get back the paragraph you deleted or the video footage you removed. For your document, you may have maintained a manual version control and listed down every possible change, or most likely not done it at all. On the other hand, you might have set up a backup to be saved automatically. Retrieving your video editing project could be even more difficult.
2. You are working on your project and your computer battery suddenly dies, or your computer freezes and needs a reboot; you could suddenly lose work and in case you forgot to save it, you may have to start all over again.

There could be a number of other problems that cause you to lose your work, resulting in wasted time, effort and frustration. So, what are your options? Storage systems such as Rockstor offer the Snapshots technology that is designed to solve problems like those described above. These offer not just a solution, but also allow you to create a strategy to proactively manage these issues before they arise.



Figure 2: Disks